[HINTS & SOLUTIONS]

Q.1 3

Sol. (x_1, y_2) lies on line

$\therefore\ 3x_1 - 4y_2 - a(a-2) = 0$

$\therefore\ 4y_2 = 3x_1 - a(a-2) = 0$

Now, $y_2 < y_1 \Rightarrow \dfrac{3x_1 - a(a-2)}{4} < y_1$

$\Rightarrow 3(2b+3) - a(a-2) < 4b^2$ Put $x_1 = 2b + 3;\ y_1 = b^2$

$\Rightarrow a^2 - 2a + 4b^2 - 6b - 9 > 0\ \forall\ a \in R$

$\Rightarrow D < 0 \Rightarrow 4 - 4(4b^2 - 6b - 9) < 0$

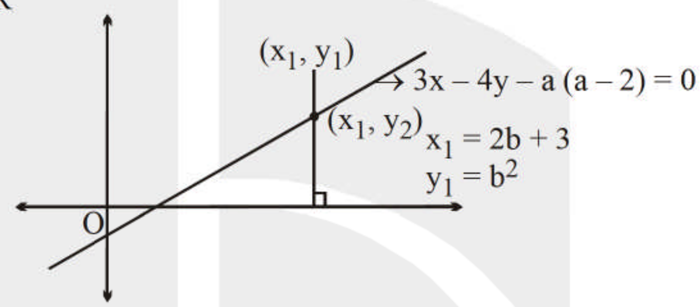
$\Rightarrow 1 - 4b^2 + 6b + 9 < 0$

$\Rightarrow 4b^2 - 6b - 10 > 0$

$\Rightarrow 2b^2 - 3b - 5 > 0$

$\Rightarrow (2b-5)(b+1) > 0$

$\Rightarrow b \in (-\infty, -1) \cup \left(\dfrac{5}{2}, \infty\right)$



Hence, least positive integral value 'b' is 3.

Aliter: $-3x + 4y + a(a-2) = 0$

$(2b + 3, b^2)$ lies above the line.

$-3(2b+3) + 4b^2 + a(a-2) > 0\ \forall\ a \in R$

$\Rightarrow a^2 - 2a + 4b^2 - 6b - 9 > 0$

$\Rightarrow D < 0 \Rightarrow 4 - 4(4b^2 - 6b - 9) < 0$

$\Rightarrow 1 - 4b^2 + 6b + 9 < 0$

$\Rightarrow 4b^2 - 6b - 10 > 0$

$\Rightarrow 2b^2 - 3b - 5 > 0$

$\Rightarrow (2b-5)(b+1) > 0 \Rightarrow b \in (-\infty, -1) \cup \left(\dfrac{5}{2}, \infty\right)$

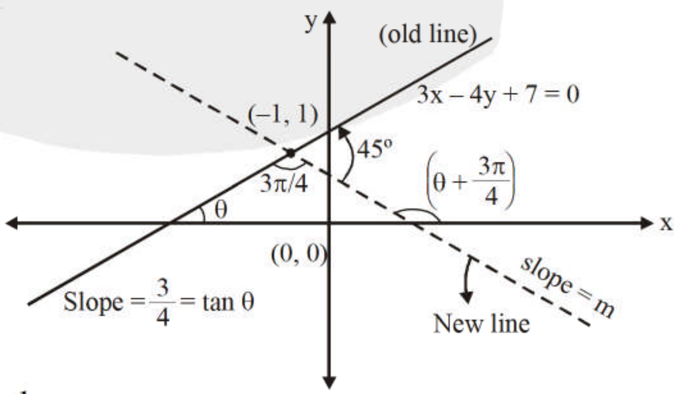
Hence, least positive integral value 'b' is 3.

Q.2 $x + 7y - 6 = 0$ (B) $5y - x - 6 = 0$

Sol. Let

$m = \text{slope of new line} = \tan\left(\theta + \dfrac{3\pi}{4}\right)$

$= \dfrac{\tan\theta - 1}{1 + \tan\theta} = \dfrac{\frac{3}{4} - 1}{1 + \frac{3}{4}} = \dfrac{\frac{-1}{4}}{\frac{7}{4}} = \dfrac{-1}{7}$



$\therefore$ The equation of line in new position is

$(y - 1) = \dfrac{-1}{7}(x + 1) \Rightarrow 7y - 7 = -x - 1$

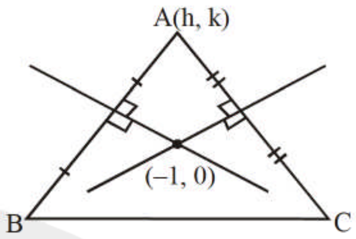
$\Rightarrow x + 7y - 6 = 0$

Aliternative :

$$\frac{\frac{3}{4}-m}{1+\frac{3m}{4}} = 1 = \tan\frac{\pi}{4} \Rightarrow m = \frac{-1}{7} \text{ (According to the direction shown)}$$

Q.3 13

Sol. The given lines intersect at (– 1, 0)



The vertex A lies on circle having centre at (– 1, 0) and radius 2 units

$\Rightarrow$ Locus of A is $(x + 1)^2 + y^2 = 4$.

Hence g = 2 and c = – 3

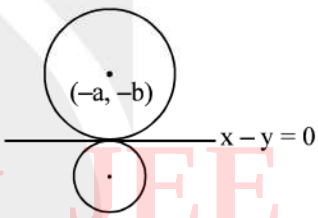
$\Rightarrow$ $g^2 + c^2 = 13$.

Q.4 1

Sol. radical axis $x - y = 0$ ....(1)

centre of 1st circle is (–a, –b) and r = $\sqrt{a^2+b^2-c}$

now perpendicular distance from centre on radical axis = radius of the circle



$$(a - b)^2 = 2[a^2 + b^2 - c]$$

$$\Rightarrow \quad a^2 + b^2 - 2ab = 2[a^2 + b^2 - c]$$

$$\Rightarrow \quad a^2 + b^2 + 2ab = 2c \quad \Rightarrow \quad \frac{(a+b)^2}{2c} = 1$$

Alternatively: $2x^2 + 2(a + b)x + c = 0$; put D = 0

$$4(a + b)^2 = 8c \quad \Rightarrow \quad \frac{(a+b)^2}{2c} = 1$$

Q.5 straight line of slope $\frac{5}{3}$

Sol. Let the circle be $x^2 + y^2 + 2gx + 2fy = 0$

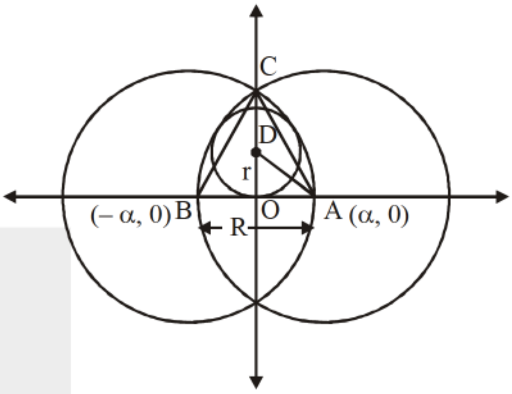
It cuts the other circle orthogonally, if

$$2g\left(\frac{-5}{2}\right) + 2f\left(\frac{3}{2}\right) = 0 + (-1)$$

Hence, the locus of the centre $(-g, -f)$ is $5x - 3y + 1 = 0$, which is a straight line of slope $\frac{5}{3}$.

Q.6 6

Sol. ΔABC is equilateral, also $\frac{2\pi R}{6} = 8\pi$



$R = 24$

In ΔAOD, $r^2 + \frac{R^2}{4} = (R - r)^2$

$$\frac{R}{4} = R - 2r$$

$$r = \frac{3R}{8} \Rightarrow r = 9 \text{ and } \alpha = 12$$

Centre of circle (0, 9)

$\Rightarrow$ $(x - 0)^2 + (y - 9)^2 = 9^2$

$\therefore$ $|a + b + r - \alpha| = |9 + 9 - 12| = 6$

Q.7 3

Sol. $\because$ $p \in I$

$\therefore$ pont is $(2p - 4, 0)$

now $(2p - 4)^2 - 9 < 0$

$\frac{1}{2} < p < \frac{7}{2}$.

Q.8 2

Sol. We have $x^2 + y^2 + \left(\frac{\lambda}{2}\right)x - \left(\frac{1+\lambda^2}{2}\right)y - 5 = 0$ ......(1)

and $x^2 + y^2 + 4x + 6y + 3 = 0$ ......(2)

As (1) and (2) cut each other orthogonally, so

$$2\left(\left(\frac{\lambda}{4}\right)\cdot 2 + \left(\frac{-1-\lambda^2}{4}\right)\cdot(3)\right) = -5 + 3$$

$\Rightarrow$ $\lambda - \frac{3(1+\lambda^2)}{2} = -2$

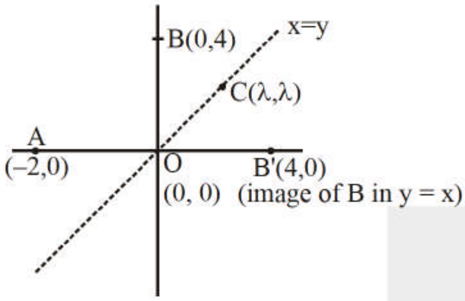
$\Rightarrow$ $2\lambda - 3 - 3\lambda^2 = -4 \Rightarrow 3\lambda^2 - 2\lambda - 1 = 0$

(As discriminant of quadratic equation in λ is positive)

So, two distinct real values of λ are possible.

Q.9 0

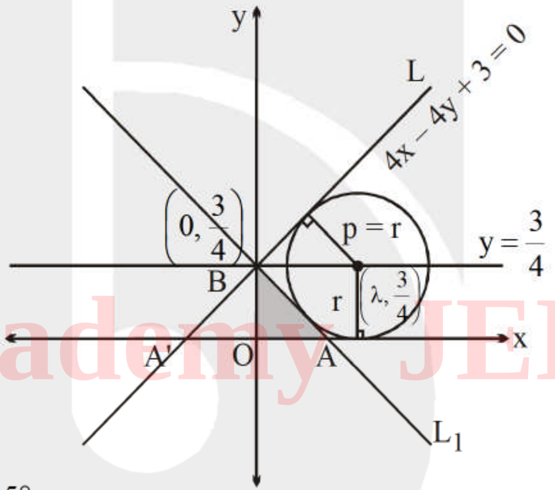
Sol. For perimeter of ΔABC to be least $(AC + BC)$ must be least hence C should be (0, 0).



Q.10 1

Q.11 41

Sol. Using p = r



(i) $\dfrac{4\lambda - 3 + 3}{4\sqrt{2}} = r = \dfrac{3}{4} \Rightarrow \dfrac{\lambda}{\sqrt{2}} = \dfrac{3}{4}$

$\therefore\ \lambda = \dfrac{3}{2\sqrt{2}} \Rightarrow [\lambda] = 1$

(ii) $\Delta OAB = \dfrac{1}{2} \times \dfrac{3}{4} \times \dfrac{3}{4} = \dfrac{9}{32} \equiv \dfrac{p}{q}$

$\therefore$ (p + q) = 41. **Ans.**

Note: slope of line L is 1, hence $\angle OA'B = 45°$

Hence slope of line L_1 is – 1.

Q.12 25

Q.13 75

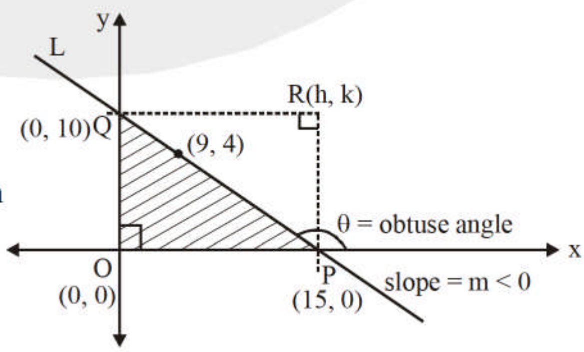
Q.14 $\dfrac{9}{x} + \dfrac{4}{y} = 1$

Sol.

(i) Let equation of line L be $(y - 4) = m(x - 9)$

$\therefore\ P\left(9 - \dfrac{4}{m}, 0\right),\ Q(0, 4 - 9m)$

Now,



$$OP + OQ = \underbrace{\left(9 - \frac{4}{m}\right)}_{\substack{\text{positive } q + y \\ \text{as } m < 0}} + \underbrace{(4 - 9m)}_{\substack{\text{positive } q + y \\ \text{as } m < 0}} = 13 - \frac{4}{m} - 9m$$

$$\therefore\ OP + OQ \geq 13 + 2\sqrt{\left(\frac{-4}{m}\right)(-9m)} = 25$$

$\Rightarrow$ Minimum value of $OP + OQ$ equal 25, when $\frac{-4}{m} = -9m \;\Rightarrow m^2 = \frac{4}{9}$

$\therefore\; m = \frac{-2}{3}$ (As $m < 0$)

(ii) So, $ar(\Delta OPQ) = \frac{1}{2}(15)(10) = 75$

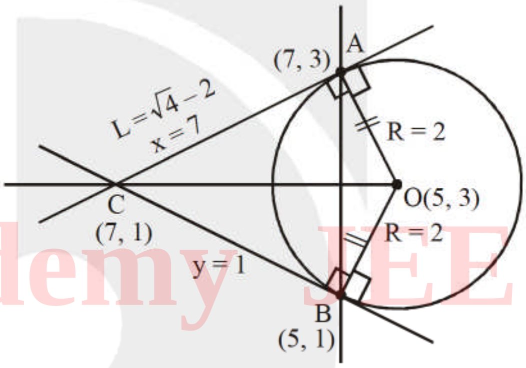
(iii) Let $R(h, k)$. So

$$h = \frac{4}{m} \quad \ldots\ldots (1) \qquad k = 4 - 9m \quad \ldots\ldots (2)$$

$\therefore$ Eliminating m locus of $R(h, k)$ is $\left(\frac{9}{x} + \frac{4}{y} = 1\right)$

Q.15 (A) area of quadrilateral OACB equals 4.
(C) the smallest possible circle of the family S = 0 is $x^2 + y^2 - 12x - 4y + 38 = 0$.
(D) the coordinates of point C are (7, 1).

Sol. Coordinates of O are (5, 3) and radius = 2.



Equation of tangent at A(7, 3) is
$$7x + 3y - 5(x + 7) - 3(y + 3) + 30 = 0$$
i.e. $x = 7$

Also, equation of tangent at B(5, 1) is
$$5x + y - 5(x + 5) - 3(y + 1) + 30 = 0$$
i.e. $y = 1$

$\therefore$ Coordinates of C are (7, 1)

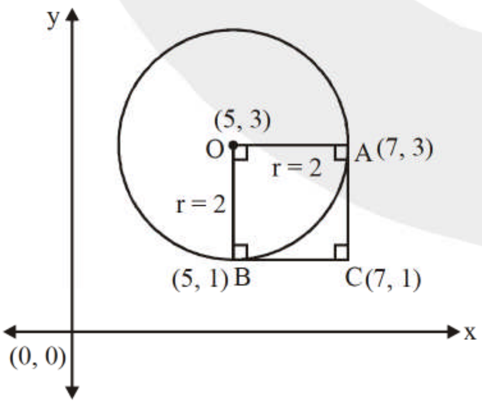
So, area of quadrilateral OACB

$$= 2\left(\frac{1}{2}LR\right) = LR = \left(\sqrt{4}\right)2 = 4$$

Also, equation of AB is $x - y = 4$ (radical axis).

Note that the equation of circle of smallest radius of family S = 0 is circle described on AB as diameter given by i.e. $x^2 + y^2 - 12x - 4y + 38 = 0$

Aliternative:



Q.16 (A) $p + q = 18$ (C) $p - q = 4\sqrt{2}$ (D) $p\,q = 73$

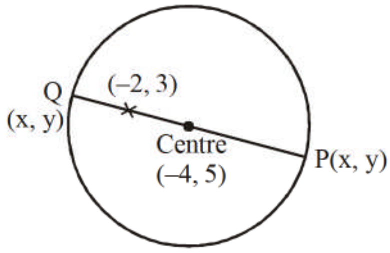
Sol. For the circle,

$$x^2 + y^2 + 8x - 10y - 40 = 0,$$

centre is $(-4, 5)$ and radius $= 9$

Also, distance of the centre $(-4, 5)$ from $(-2, 3) = \sqrt{4+4} = 2\sqrt{2} < 9$

So, $(-2, 3)$ lies inside the given circle



$\therefore\ p = \max \cdot \sqrt{(x+2)^2 + (y-3)^2} = 9 + 2\sqrt{2}$

& $q = \min \cdot \sqrt{(x+2)^2 + (y-3)^2} = 9 - 2\sqrt{2}$

So, $p + q = 18$

$p - q = 4\sqrt{2}$

Also, $pq = 81 - 8 = 73$

Q.17 (C) concurrent for no value of k (D) parallel for $k = 3$

Sol. For concurrency of 3 lines, put $\Delta = 0$

$$\Rightarrow \begin{vmatrix} 1 & 1 & -1 \\ k-1 & k^2-7 & -5 \\ k-2 & 2k-5 & 0 \end{vmatrix} = 0$$

$\Rightarrow\ k^3 - 4k^2 + 5k - 6 = 0$

$\Rightarrow\ (k - 3)(k^2 - k + 2) = 0$

$\Rightarrow\ k = 3$ but $k^2 - k + 2 = 0$ has no real roots. Also, for $k = 3$, the lines are parallel.

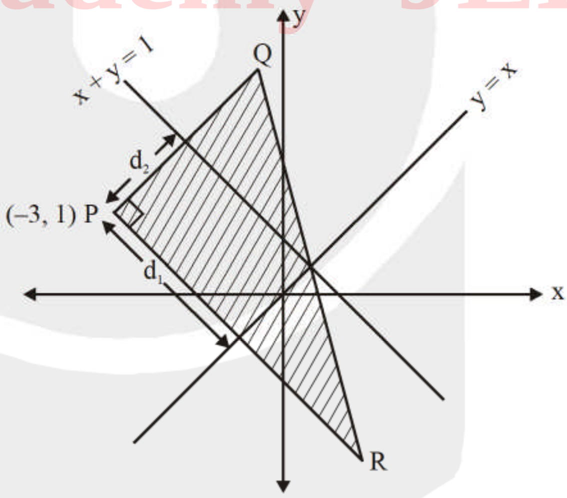
Q.18 3

Sol. $x^2 - y^2 - (x - y) = 0$

$(x - y)(x + y - 1) = 0$

$$d_1 = \left|\frac{-3-1}{\sqrt{2}}\right| = 2\sqrt{2}$$

$$d_2 = \left|\frac{-3+1-1}{\sqrt{2}}\right| = \frac{3}{\sqrt{2}}$$



$\therefore$ Area of ΔPQR,

$$S = \frac{1}{2} \times 4\sqrt{2} \times 3\sqrt{2} \ \Rightarrow\ S = 12$$

$$\therefore\ \left[\sqrt{S}\right] = 3$$

Aliter: $P \equiv (-3, 1)$; $Q \equiv (1, -3)$; $R \equiv (1, 3)$

$$S = \frac{1}{2}\begin{vmatrix} -3 & 1 & 1 \\ 1 & -3 & 1 \\ 1 & 3 & 1 \end{vmatrix} = 12$$

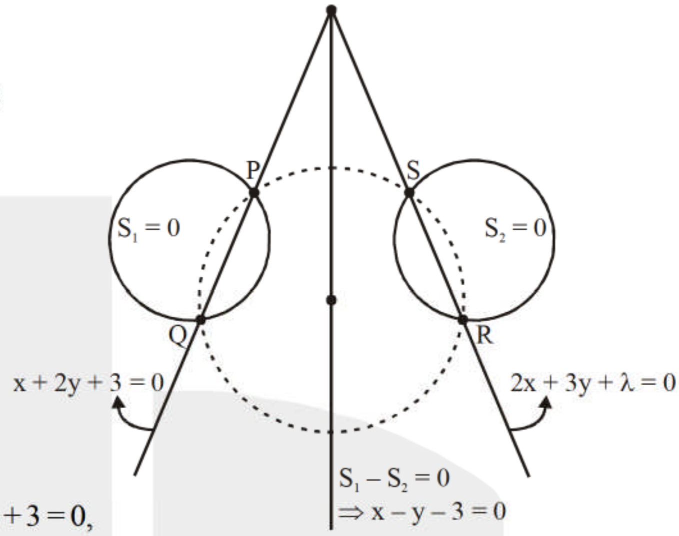
Q.19 $\lambda = 4$

Sol. As, lines

$$2x + 3y + \lambda = 0,\ x + 2y + 3 = 0$$

and $x - y - 3 = 0$ are concurrent,

As we know that the radical axis of 3 circles taken two at a time are concurrent.



So $\begin{vmatrix} 2 & 3 & \lambda \\ 1 & 2 & 3 \\ 1 & -1 & -3 \end{vmatrix} = 0$

$\Rightarrow\ 2(-6 + 3) - 3\,(-6) + \lambda(-3) = 0$

$\Rightarrow\ 12 = 3\lambda$

$\Rightarrow\ \lambda = 4$ **Ans.**

Aliternative :

Family of circle passing through line $x + 2y + 3 = 0$,

circle $x^2 + y^2 + 4x + 4y - 1 = 0$, is

$$(x^2 + y^2 + 4x + 4y - 1) + t(x + 2y + 3) = 0$$

i.e. $x^2 + y^2 + (4 + t)x + (4x + 2t)y + (3t - 1) = 0$

Similarly, family of through line $2x + 3y + \lambda = 0$, circle $(x^2 + y^2 + 6x + 2y - 7) + s(2x + 3y + \lambda) = 0$

i.e. $x^2 + y^2 + (6 + 28)x + (2 + 3s)y + (\lambda s - 7) = 0$ ......(2)

Now, (1) and (2) represent the same circle,

So, $(4 + t) = 6 + 2s$

$(4 + 2t) = 2 + 3s$

$\therefore$ on solving, we get $s = -6,\ t = -10$

Also, $\lambda s = 7 = 3t - 1 \Rightarrow \lambda(-6) - 7 = 3\,(-10) - 1$

$\Rightarrow\ 6\lambda = 24 \quad \Rightarrow \quad \lambda = 4$